

Geometrical Linear Transformations in 2D and 3D

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Preliminaries

- Systems of Linear Equations, Row-Echelon Forms, Elementary Row Operations, Gauss-Jordan Elimination
- Matrix Algebra and Properties, Inverse Matrices, Properties of Determinant
- Euclidean Vector Spaces, Inner Product, L^2 norm, Distance and Angle between Vectors
- The above topics should've been covered in the early weeks of MA2001/MA1522/MA1508E. We will not go over them here, but understanding these properties well is important for understanding these notes and the applications of Linear Algebra (LA).
- The topics that should've been covered in LA classes that we will go over again briefly here include linear spans and basis, dimension, rank and kernel, orthogonality, eigenanalysis and linear transformations. The goal here is to provide geometric intuition rather than mathematical rigour.
- Note that decompositions such as LU, QR and SVD find uses in other applied LA fields, and SVD in particular does find a use in 3D geometry (finding the best fit plane for a set of points in 3D). However, these would otherwise be considered secondary for basic concepts in the geometry of Euclidean vector spaces.

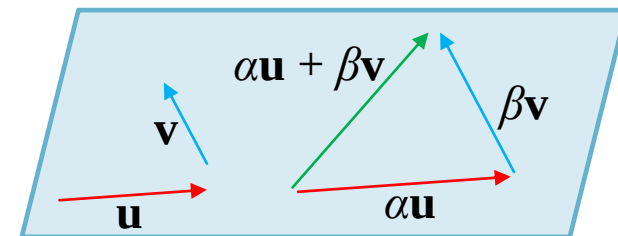
Recap: Linear combinations and spans

- Let $S = \{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_k\}$ be a set of vectors in $\mathbb{R}^n$. A linear combination of these vectors is $c_1\mathbf{u}_1 + c_2\mathbf{u}_2 + \dots + c_k\mathbf{u}_k$, for some constants $c_1, c_2, \dots, c_k \in \mathbb{R}$.
- The span of S is the subset of $\mathbb{R}^n$ containing all the linear combinations of the vectors in S . That is, given $\mathbf{v} \in \mathbb{R}^n$,

$$\mathbf{v} \in \text{span}(S) \Leftrightarrow \exists c_1, c_2, \dots, c_k \in \mathbb{R} \{ \mathbf{v} = c_1\mathbf{u}_1 + c_2\mathbf{u}_2 + \dots + c_k\mathbf{u}_k \}.$$

- For example, let $S = \left\{ \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \right\}$. We have $\begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} \in \text{span}(S)$, but $\begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \notin \text{span}(S)$.
- Form the matrix $\mathbf{A} = (\mathbf{u}_1 \ \mathbf{u}_2 \ \dots \ \mathbf{u}_k)$. Notice that if the system $\mathbf{A}\mathbf{x} = \mathbf{v}$ is consistent (where $\mathbf{x} \in \mathbb{R}^k$), then $\mathbf{v} \in \text{span}(S)$. Intuitively, we are looking for the constants $c_1, c_2, \dots, c_k$ to find a linear combination of vectors in S that gives $\mathbf{v}$.
- The converse is also true: If $\mathbf{v} \in \text{span}(S)$, then $\mathbf{A}\mathbf{x} = \mathbf{v}$ must be consistent.
- Spans are closed under linear combination. That is,

$$\mathbf{u}, \mathbf{v} \in \text{span}(S) \wedge \alpha, \beta \in \mathbb{R} \Rightarrow \alpha\mathbf{u} + \beta\mathbf{v} \in \text{span}(S).$$



- Geometrically, the region shown is $\text{span}\{\mathbf{u}, \mathbf{v}\}$. Any linear combination of $\mathbf{u}$ and $\mathbf{v}$ lies in the span.

Questions

Suppose $\mathbf{v}$ is a linear combination of the vectors in $S = \{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_k\}$.

- Is the linear combination for $\mathbf{v}$ necessarily unique?
- Show that if $\mathbf{A} = (\mathbf{u}_1 \ \mathbf{u}_2 \ \dots \ \mathbf{u}_k)$ and $\text{RREF}(\mathbf{A})$ has no zero rows, then $\text{span}(S) = \mathbb{R}^n$.

Answer:

- 1) If $\mathbf{0} \notin S$ but $c_1\mathbf{u}_1 + c_2\mathbf{u}_2 + \dots + c_k\mathbf{u}_k = \mathbf{0}$ and at least one of $c_1, c_2, \dots, c_k$ is non-zero, then linear combinations are not necessarily unique.
- 2) This is equivalent to the system $\mathbf{Ax} = \mathbf{v}$ being consistent for all $\mathbf{v} \in \mathbb{R}^n$.

Recap: Linear spaces (subspaces)

- A linear space (subspace) over n -dimensional real vectors is a space in which vectors are closed in the space under addition and scalar multiplication. That is, if S is a linear space and $\mathbf{a}_1, \mathbf{a}_2 \in S$, then we must have $\mathbf{a}_1 + \mathbf{a}_2 \in S$ and $c\mathbf{a}_1 \in S$, where $c \in \mathbb{R}$.
- An important corollary is that any linear space must contain the zero vector.
- Linear spaces can also be defined over $n \times m$ matrices, but we will rarely play around with these cases.
- An equivalent definition is as follows: A subset $V \subseteq \mathbb{R}^n$ is a subspace if and only if it is a linear span, $V = \text{span}(S)$, for some finite set $S \{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_k\}$.
- To show that a subset V is a linear space, we can either find a spanning set, or show that V contains the zero vector and is closed under linear combination.
- Question: When is V not a linear space?

Geometrical interpretation of linear spaces

- The space $\{(x, y, z) : x + y + z = 0\}$ is a linear space, but the space $\{(x, y, z) : x(y + z) = 0\}$ is not, since the space is not closed under addition; $(1, 0, 0)$ and $(0, 1, 0)$ are both in the space, but their sum $(1, 1, 0)$ is not.
- Geometrically, the linear space $\{(x, y, z) : x + y + z = 0\}$ is the plane with a normal vector parallel to $(1, 1, 1)$ and passing through the origin.
- Similarly, the linear space $\{(x, y, z) : x = y = z\}$ is geometrically represented by the line with a direction vector parallel to $(1, 1, 1)$ and passing through the origin.
- The definition of the latter space is actually just a shorthand for the system $\{1x - 1y + 0z = 0, 1x + 0y - 1z = 0\}$. That is, it is the intersection of two non-parallel planes, which as one can recall from A-level maths, is a line.
- The space $\{\mathbf{0}\}$ is also a linear space. It represents the origin!
- $\mathbb{R}^3$ is also a linear space. It represents the entire space of our 3D world!

Affine spaces

- An affine space can be thought of as a ‘translated’ linear space.
- The subset $S = \{(x, y, z) : x + y + z = 1\}$ is not a linear space since it does not contain $\mathbf{0}$.
- However, let $S' = \{(x, y, z) : x + y + z = 0\}$. Check that S' is a linear space. Notice also that $(0, 0, 1) \in S$, and that for all $\mathbf{v} \in S$, $\mathbf{v} - (0, 0, 1) \in S'$.
- More generally, a subset S is an affine space if there exists a linear space S' and a constant vector $\mathbf{v}_0 \in S$ such that for all $\mathbf{v} \in S$, $\mathbf{v} - \mathbf{v}_0 \in S'$. If $\mathbf{v}_0 = \mathbf{0}$, then S is itself a linear space and we have $S \subseteq S'$.
- Geometrically, the plane $x + y + z = 1$ is a translation of the plane $x + y + z = 0$.
- Affine spaces generalise linear spaces in a way that is useful for computer graphics and other applications. The gist of it is that while linear transformations are forced to be ‘centred’ at the origin, affine transformations can be used to transform a set of vectors with a ‘centre’ wherever we want.

Recap: Basis, Dimension, Column / Row Spaces, Null Spaces, Rank / Nullity

- A basis B for a linear space S is such that B consists of linearly independent vectors and $\text{span}(B) = S$.
- The dimension of S is the number of vectors in B . Any basis must have the same dimension d .
 - If a set has fewer than d vectors, it cannot span S .
 - If a set has more than d vectors, it must be linearly dependent.
- The column or row space of a $m \times n$ matrix $\mathbf{A}$ is the subspace of $\mathbb{R}^m$ or $\mathbb{R}^n$ spanned by the columns or rows of $\mathbf{A}$.
- The null space of a $m \times n$ matrix $\mathbf{A}$ is the subspace of $\mathbb{R}^n$ containing all vectors $\mathbf{v}$ which satisfy $\mathbf{A}\mathbf{v} = \mathbf{0}$.
- The rank of a matrix is the number of linearly independent rows or columns (or equivalently, the dimension of its column space or row space), and the nullity of a matrix is the dimension of its null space.
- The rank-nullity theorem asserts that $\text{rank}(\mathbf{A}) + \text{nullity}(\mathbf{A}) = n$.
- Geometrically, the dimension of a subspace is the number of linearly independent vectors needed to specify any point in a given space. Dimension 1 is a line, dimension 2 is a plane, etc.
- While we will not cover any immediate geometrical implications of the rank-nullity theorem, it will be needed for a proof later. Do refer to the MA2001/MA1522 lecture notes for the full details if you need a refresher.

Recap: Orthogonality

- Orthogonality is a generalisation of the concept of perpendicularity in 2D and 3D, just as linear independence is a generalisation of the concept of parallelism.
- Two vectors are said to be orthogonal to each other if and only if their inner product is zero.
- A set is orthogonal if its vectors are all pairwise orthogonal, and a set is orthonormal if it is orthogonal and the vectors all have unit norm.
- A vector $\mathbf{u}$ is orthogonal to a subspace V if $\mathbf{u} \cdot \mathbf{v} = 0$ for every $\mathbf{v} \in V$. Alternatively, if S is a spanning set for V , then we have $\mathbf{u} \perp V \Leftrightarrow \mathbf{u} \in \text{Null}(\mathbf{A}^T)$, where $\mathbf{A}$ is the matrix whose columns are the column vectors in S .
- The orthogonal complement of V , denoted $V^\perp$, is the set of all vectors that are orthogonal to V . A theorem is that if $\mathbf{A}$ is the matrix defined above, then $V^\perp = \text{Null}(\mathbf{A}^T)$.
- An orthogonal (orthonormal) basis is a basis with orthogonal (orthonormal) vectors.
- A square matrix $\mathbf{M}$ is orthogonal if and only if its columns (or rows) form an orthonormal basis, or equivalently, $\mathbf{M}^T = \mathbf{M}^{-1}$.

Recap: Orthogonal Projection and the Gram-Schmidt Process

- The orthogonal projection theorem states that given a subspace V of $\mathbb{R}^n$, every vector in $\mathbb{R}^n$ can be decomposed *uniquely* as a sum of two vectors, one in V and one in $V^\perp$.
- Specifically, if $\{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_k\}$ is an orthogonal basis for V , and $\mathbf{w} = \mathbf{w}_p + \mathbf{w}_n$ where $\mathbf{w}_p \in V$, $\mathbf{w}_n \in V^\perp$, then

$$\mathbf{w}_p = \frac{\mathbf{w} \cdot \mathbf{u}_1}{\mathbf{u}_1 \cdot \mathbf{u}_1} \mathbf{u}_1 + \frac{\mathbf{w} \cdot \mathbf{u}_2}{\mathbf{u}_2 \cdot \mathbf{u}_2} \mathbf{u}_2 + \dots + \frac{\mathbf{w} \cdot \mathbf{u}_k}{\mathbf{u}_k \cdot \mathbf{u}_k} \mathbf{u}_k.$$

- The Gram-Schmidt process converts a linearly independent set to an orthogonal set. It proceeds as follows:

$$\mathbf{v}_1 = \mathbf{u}_1$$

$$\mathbf{v}_2 = \mathbf{u}_2 - \left(\frac{\mathbf{v}_1 \cdot \mathbf{u}_2}{\|\mathbf{v}_1\|^2} \right) \mathbf{v}_1$$

$$\mathbf{v}_3 = \mathbf{u}_3 - \left(\frac{\mathbf{v}_1 \cdot \mathbf{u}_3}{\|\mathbf{v}_1\|^2} \right) \mathbf{v}_1 - \left(\frac{\mathbf{v}_2 \cdot \mathbf{u}_3}{\|\mathbf{v}_2\|^2} \right) \mathbf{v}_2$$

$$\vdots$$

$$\mathbf{v}_k = \mathbf{u}_k - \left(\frac{\mathbf{v}_1 \cdot \mathbf{u}_k}{\|\mathbf{v}_1\|^2} \right) \mathbf{v}_1 - \left(\frac{\mathbf{v}_2 \cdot \mathbf{u}_k}{\|\mathbf{v}_2\|^2} \right) \mathbf{v}_2 - \dots - \left(\frac{\mathbf{v}_{k-1} \cdot \mathbf{u}_k}{\|\mathbf{v}_{k-1}\|^2} \right) \mathbf{v}_{k-1}.$$

Recap: Eigenvalues and eigenvectors

- Given a square matrix $\mathbf{A}$ of order n and a non-zero vector $\mathbf{v} \in \mathbb{C}^n$, we say that $\mathbf{v}$ is an eigenvector of $\mathbf{A}$ if there exists $\lambda \in \mathbb{C}$ such that $\mathbf{A}\mathbf{v} = \lambda\mathbf{v}$. In such a case, λ is known as an eigenvalue of $\mathbf{A}$.
- For example, consider the matrix $\mathbf{A} = \begin{pmatrix} 0 & 1 \\ -2 & 3 \end{pmatrix}$. Observe that $\begin{pmatrix} 0 & 1 \\ -2 & 3 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$ and $\begin{pmatrix} 0 & 1 \\ -2 & 3 \end{pmatrix} \begin{pmatrix} 1 \\ 2 \end{pmatrix} = \begin{pmatrix} 2 \\ 4 \end{pmatrix} = 2\begin{pmatrix} 1 \\ 2 \end{pmatrix}$. Hence, 1 and 2 are eigenvalues of $\mathbf{A}$ with respective eigenvectors $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$ and $\begin{pmatrix} 1 \\ 2 \end{pmatrix}$.
- Eigenvectors are not unique as any non-zero scalar multiple of an eigenvector is also an eigenvector.
- Furthermore, it is possible that multiple linearly independent vectors are eigenvectors of a square matrix which correspond to the same eigenvalue. Consider $\mathbf{I}_2 \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$, and $\mathbf{I}_2 \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$. Hence, $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $\begin{pmatrix} 0 \\ 1 \end{pmatrix}$ are eigenvectors of $\mathbf{I}_2$, and both have corresponding eigenvalue 1.
- Note here that although we will only discuss matrices whose entries are all in $\mathbb{R}$, it is entirely possible that their eigenvalues and eigenvectors are in $\mathbb{C}$.

Recap: Eigenvalues and eigenvectors

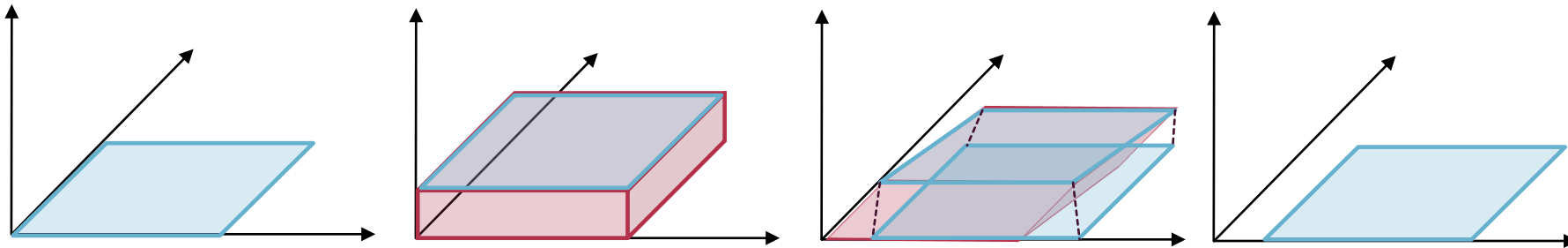
- Given a square matrix $\mathbf{A}$ of order n and a non-zero vector $\mathbf{v} \in \mathbb{C}^n$, we say that $\mathbf{v}$ is an eigenvector of $\mathbf{A}$ if there exists $\lambda \in \mathbb{C}$ such that $\mathbf{A}\mathbf{v} = \lambda\mathbf{v}$. In such a case, λ is known as an eigenvalue of $\mathbf{A}$.
- An equivalent statement to the above is that $(\lambda\mathbf{I} - \mathbf{A})\mathbf{v} = \mathbf{0}$, that is, $\mathbf{v}$ is a non-trivial solution of the homogenous system of equations $(\lambda\mathbf{I} - \mathbf{A})\mathbf{x} = \mathbf{0}$.
- Hence, the set of all solutions to the above system forms a subspace of $\mathbb{C}^n$. This subspace is also known as the eigenspace of $\mathbf{A}$ associated to λ .
- Question: Can 0 be an eigenvalue of $\mathbf{A}$? What would it imply?

Introduction to Homogenous Coordinates: Translation of a point in 2D

- Consider a transformation which translates a point $P : (p_1, p_2)$ in the xy -plane to the point $P' : (p_1 + a_1, p_1 + a_2)$. That is, we want a translation from P to P' .
- Recall that a transformation T is linear only if $T(\mathbf{0}) = \mathbf{0}$. Hence, there is no linear transformation $\mathbb{R}^2 \rightarrow \mathbb{R}^2$ that maps the position vector of P to the position vector of P' unless $a_1 = a_2 = 0$, which is just the identity transformation.
- However, what if we move P to a 3D space? Here's the idea: We can temporarily add a third dimension so that we can map p_1 to $p_1 + a_1$ and p_2 to $p_2 + a_2$. In order to achieve this, we can add a_1 and a_2 to the third column of our transformation matrix, and make the third row of the matrix $(0 \ 0 \ 1)$, to ensure the z -coordinate remains unchanged.
- There exists a linear transformation $T : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ such that $T\left(\begin{pmatrix} p_1 \\ p_2 \\ 1 \end{pmatrix}\right) = \begin{pmatrix} p_1 + a_1 \\ p_2 + a_2 \\ 1 \end{pmatrix}$. The matrix $\begin{pmatrix} 1 & 0 & a_1 \\ 0 & 1 & a_2 \\ 0 & 0 & 1 \end{pmatrix}$ is its standard matrix. Check that, for any point in $\mathbb{R}^3$, the z -coordinate remains unchanged when T is applied to it.
- If we want to compose multiple 2D affine transformations, we can chain up multiple 3×3 matrices representing linear transformations with homogenous coordinates. Notice that only the x and y -coordinates will change due to these transformations, while the z -coordinate will always remain constant.

Representing Affine Transformations as Linear Transformations using Homogenous Coordinates

- We have just seen that a square matrix of order 3 can be used to represent translations in $\mathbb{R}^2$. Geometrically, a translation in an $\mathbb{R}^2$ space can be achieved by extruding the space into $\mathbb{R}^3$ using prism extrusion, followed by a shearing in this new $\mathbb{R}^3$ space, followed by a projection back into the original $\mathbb{R}^2$ space, as shown below:



- In general, for a translation in $\mathbb{R}^n$, an $(n + 1)$ -dimensional space is needed to represent this translation as a linear transformation. We say that translation in $\mathbb{R}^n$ is an *affine transformation* in $\mathbb{R}^n$. All linear transformations are affine transformations, but not all affine transformations are linear transformations.
- More generally, if $T : \mathbb{R}^n \rightarrow \mathbb{R}^n$ is an affine transformation, it can be expressed as the composition of a translation and a linear transformation, and there exists a linear transformation $T' : \mathbb{R}^{n+1} \rightarrow \mathbb{R}^{n+1}$ such that $T'\left(\begin{pmatrix} \mathbf{p} \\ 1 \end{pmatrix}\right) = \begin{pmatrix} T(\mathbf{p}) \\ 1 \end{pmatrix}$ for any vector $\mathbf{p} \in \mathbb{R}^n$.

Reflection of a point about a line in 2D/3D

- Consider the line $L : \mathbf{r} = \mathbf{a} + \lambda \mathbf{d}$, where $\lambda \in \mathbb{R}$ and $\mathbf{d} \neq \mathbf{0}$, and the point P with position vector $\mathbf{p}$ in the xyz -space, and let P' denote the reflection of P in L . What is the position vector of P' ?
- This is of course a standard A-level problem. Assuming $\mathbf{f}$ is the position vector of the foot of perpendicular from P to L , we know that $\mathbf{p}' = 2\mathbf{f} - \mathbf{p}$. We also know using the projection formula that $\mathbf{f} = \frac{((\mathbf{p} - \mathbf{a}) \cdot \mathbf{d})\mathbf{d}}{\|\mathbf{d}\|^2} + \mathbf{a}$.
- For the 3D case, if we let $\mathbf{a} = \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix}$, and denote $\mathbf{d}$ and $\mathbf{p}$ similarly, the position vector of $\mathbf{p}'$ can be expressed as
$$\begin{pmatrix} (cd_1^2 - 1)p_1 + cd_1d_2p_2 + cd_1d_3p_3 + 2a_1 - cd_1(a_1d_1 + a_2d_2 + a_3d_3) \\ cd_1d_2p_1 + (cd_2^2 - 1)p_2 + cd_2d_3p_3 + 2a_2 - cd_2(a_1d_1 + a_2d_2 + a_3d_3) \\ cd_1d_3p_1 + cd_2d_3p_2 + (cd_3^2 - 1)p_3 + 2a_3 - cd_3(a_1d_1 + a_2d_2 + a_3d_3) \end{pmatrix}, \text{ where } c = \frac{2}{d_1^2 + d_2^2 + d_3^2}.$$
- The 2D case should be easily deducible.
- Check that there is no linear transformation $\mathbb{R}^3 \rightarrow \mathbb{R}^3$ that maps $\mathbf{p}$ to $\mathbf{p}'$ unless $\mathbf{a} = k\mathbf{d}$ for some constant $k \in \mathbb{R}$.

Reflection of a point about a line in 2D/3D

- Using homogenous coordinates, we obtain
$$\begin{pmatrix} cd_1^2 - 1 & cd_1d_2 & cd_1d_3 & 2a_1 - cd_1(a_1d_1 + a_2d_2 + a_3d_3) \\ cd_1d_2 & cd_2^2 - 1 & cd_2d_3 & 2a_2 - cd_2(a_1d_1 + a_2d_2 + a_3d_3) \\ cd_1d_3 & cd_2d_3 & cd_3^2 - 1 & 2a_3 - cd_3(a_1d_1 + a_2d_2 + a_3d_3) \\ 0 & 0 & 0 & 1 \end{pmatrix}$$
 for the 3D case and
$$\begin{pmatrix} bd_1^2 - 1 & bd_1d_2 & 2a_1 - bd_1(a_1d_1 + a_2d_2) \\ bd_1d_2 & bd_2^2 - 1 & 2a_2 - bd_2(a_1d_1 + a_2d_2) \\ 0 & 0 & 1 \end{pmatrix}$$
 for the 2D case, where $b = \frac{2}{d_1^2 + d_2^2}$.
- However, in computer graphics it is common to set $\|\mathbf{d}\| = 1$ for computational simplicity. For the special case $\mathbf{a} = \mathbf{0}$ and $\|\mathbf{d}\| = 1$, we can rewrite the 2D case in standard coordinates as the 2×2 matrix
$$\begin{pmatrix} 2d_1^2 - 1 & 2d_1d_2 \\ 2d_1d_2 & 2d_2^2 - 1 \end{pmatrix}.$$
- The latter matrix is known as the 2×2 Householder matrix, and represents a Householder transformation, which is the reflection of a vector in $\mathbb{R}^n$ about some subspace of $\mathbb{R}^n$ with dimension $(n - 1)$.

Geometrical interpretation of eigenvalues and eigenvectors: Reflection about a line in 2D

- The determinant of a translation matrix is always 1, so we can ignore the effects of translation for eigenanalysis. This applies in general, so most affine transformations have the same determinant as their linear equivalents.
- Consider the reflection matrix $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$. Its characteristic equation is $\lambda^2 - 1 = 0$, so the eigenvalues are 1 and -1 . Corresponding eigenvectors are $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$ and $\begin{pmatrix} -1 \\ 1 \end{pmatrix}$. However, what is the actual geometric meaning?
- We have $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} y \\ x \end{pmatrix}$. The transformation $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$, given by $T((x, y)) = (y, x)$, is a linear transformation (check it!) and its standard matrix is $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$. In particular, $T(\mathbf{v})$ is a reflection of the vector $\mathbf{v}$ about the line $y = x$.
- The eigenvalue of 1 and corresponding eigenvector $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$ indicates that any vector in $\text{span}\left\{\begin{pmatrix} 1 \\ 1 \end{pmatrix}\right\}$ is invariant under T (geometrically, a reflection of a line in itself should be the same line).
- The eigenvalue of -1 and corresponding eigenvector $\begin{pmatrix} -1 \\ 1 \end{pmatrix}$ indicates that any vector in $\text{span}\left\{\begin{pmatrix} -1 \\ 1 \end{pmatrix}\right\}$ maps to its negative under T (geometrically, a reflection of L_2 in L_1 , where L_2 is perpendicular to L_1 , should just be L_2 in the opposite direction).

Exercise

- Find the eigenvalues of the matrix $\mathbf{M} = \frac{1}{49} \begin{pmatrix} +41 & -12 & -24 \\ -12 & +31 & -36 \\ -24 & -36 & -23 \end{pmatrix}$ as well as corresponding eigenbases.

One possible answer: Eigenvalue 1 with basis $\left\{ \begin{pmatrix} 0 \\ 2 \\ -1 \end{pmatrix}, \begin{pmatrix} 3 \\ 0 \\ -1 \end{pmatrix} \right\}$, eigenvalue -1 with basis $\left\{ \begin{pmatrix} 2 \\ 3 \\ 6 \end{pmatrix} \right\}$

Exercise

- Hence, identify the linear transformation represented by $\mathbf{M}$, and explain the geometric interpretation of the eigenvalues and corresponding eigenbases.

Answer: $\mathbf{M}$ represents a reflection about the plane $2x + 3y + 6z = 0$. The eigenbasis corresponding to eigenvalue 1 specifies two linearly independent directions which are invariant under the reflection, and the eigenbasis corresponding to eigenvalue -1 specifies the normal to the plane. Indeed, the cross product of the two direction vectors should produce a vector parallel to the normal vector.

Reflection of a point about a plane in 3D

- Again, this problem can be solved by any good A-level student. Here we will just state the result: To reflect the point $P : (p_1, p_2, p_3)$ in the plane $\Pi : ax + by + cz = d$, the transformation matrix using homogenous coordinates is:

$$\frac{1}{a^2 + b^2 + c^2} \begin{pmatrix} b^2 + c^2 - a^2 & -2ab & -2ac & 2ad \\ -2ab & a^2 + c^2 - b^2 & -2bc & 2bd \\ -2ac & -2bc & a^2 + b^2 - c^2 & 2cd \\ 0 & 0 & 0 & a^2 + b^2 + c^2 \end{pmatrix}$$

- Or if we assume that $\left\| \begin{pmatrix} a \\ b \\ c \end{pmatrix} \right\| = 1$:
$$\begin{pmatrix} 1 - 2a^2 & -2ab & -2ac & 2ad \\ -2ab & 1 - 2b^2 & -2bc & 2bd \\ -2ac & -2bc & 1 - 2c^2 & 2cd \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

- Again, in the special case $d = 0$, the latter matrix can be written in standard coordinates as a 3×3 matrix by omitting the last row and last column. This 3×3 matrix would also be a Householder matrix.

Exercise

Without using the general formula shown earlier, find the equation of the plane P such that a reflection in P is given

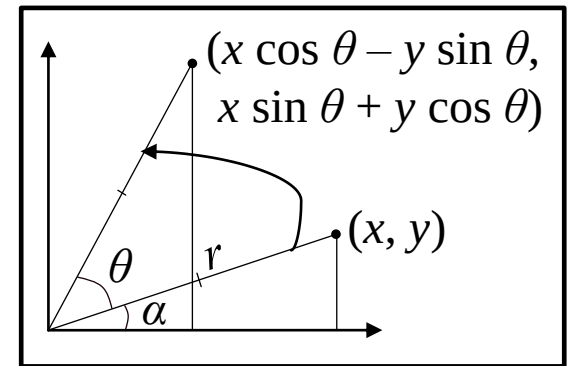
by the matrix $\mathbf{M} = \begin{pmatrix} +0.64 & -0.48 & -0.60 & +0.72 \\ -0.48 & +0.36 & -0.80 & +0.96 \\ -0.60 & -0.80 & +0.00 & +1.20 \\ +0.00 & +0.00 & +0.00 & +1.00 \end{pmatrix}$.

Trace and Determinant of Square Transformation Matrices

- Recall that for any square matrix $\mathbf{A}$, the sum of its eigenvalues is $\text{tr}(\mathbf{A})$, and the product of eigenvalues is $\det(\mathbf{A})$.
- These results follow readily from the fact that any square matrix is similar to its Jordan form (for diagonalisable matrices, the Jordan form is simply the usual diagonal matrix of eigenvalues). External reading is advised for those interested in the details.
- This has some implications for certain transformation matrices.
- Question: What are the determinants of the following matrices representing affine transformations?
 - (a) The 3×3 matrix representing a reflection about a line in 2D
 - (b) The 4×4 matrix representing a reflection about a line in 3D
 - (c) The 4×4 matrix representing a projection onto a plane in 3D

Introduction to Rotations

- Consider rotating the point P in the xy -plane by an angle θ about the origin.
- This is a linear transformation that maps (x, y) to $(x \cos \theta - y \sin \theta, x \sin \theta + y \cos \theta)$.
- To see this, set $x = r \cos \alpha$ and $y = r \sin \alpha$, where $r^2 = x^2 + y^2$ and $\tan \alpha = \frac{y}{x}$, and use the trigonometric addition formulae to expand and simplify $[r \cos(\alpha + \theta), r \sin(\alpha + \theta)]$.
- The standard matrix is $\mathbf{R}(\theta) = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$.
- The eigenvalues of $\mathbf{R}(\theta)$ satisfy the equation $\lambda^2 - (2 \cos \theta)\lambda + 1 = 0$, that is, $\lambda = \cos \theta \pm i \sin \theta$.
- Except when θ is a multiple of 180° (including 0°), the eigenvalues of $\mathbf{R}(\theta)$ are complex numbers! However, the trace of $\mathbf{R}(\theta)$ is always $2 \cos \theta$, which we can use to deduce the angle together with whether the sine of this angle is positive or negative.
- Rotations in 3D are a little more complicated, as we will see in the following pages.
- As with other linear transformations, using homogenous coordinates, we can represent an affine rotation as a composition of a translation and a linear rotation. This is true for both 2D and 3D rotations, but we will consider only the linear cases for simplicity.



Basic 3D Rotations

- We have just seen 2D rotations. We can also consider rotations in 3D.
- The xy -plane has been extruded into a 3D space by including a perpendicular z -axis. Hence, any point now has a z -coordinate, and so we can think of any rotation within the plane parallel to the xy -plane, that is, perpendicular to the z -axis. Here, we call the z -axis the *angle of rotation*.
- Similar logic applies to the x -axis and the y -axis. Since one coordinate always remains invariant under these three basic rotations, the standard rotation matrices are:

$$\mathbf{R}_x(\theta) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos \theta & -\sin \theta \\ 0 & \sin \theta & \cos \theta \end{pmatrix}, \mathbf{R}_y(\theta) = \begin{pmatrix} \cos \theta & 0 & \sin \theta \\ 0 & 1 & 0 \\ -\sin \theta & 0 & \cos \theta \end{pmatrix}, \mathbf{R}_z(\theta) = \begin{pmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

- Note that the sine terms are in alternating signs because of the complications of coupling a right-handed coordinate system with a Cartesian (x, y, z) coordinate convention.
- Recall from A level math (or physics) that $\mathbf{i} \times \mathbf{j} = \mathbf{k}$, $\mathbf{j} \times \mathbf{k} = \mathbf{i}$ and $\mathbf{k} \times \mathbf{i} = \mathbf{j}$. While the xy -plane and yz -plane have their coordinates specified in the correct order, the xz -plane has the opposite order (since $\mathbf{i} \times \mathbf{k} = -\mathbf{j}$), and so the rotation is by the angle $-\theta$ instead.

General 3D Rotations

- Intuitively, we can compose basic rotations to form general rotations. Indeed, a common way to do so is to rotate about the coordinate axes in the order of the x -axis, y -axis and z -axis. The matrix multiplication is as follows:
- Conversely, given any rotation matrix, we can solve for the angles α , β and γ according to the above convention.
- However, it's usually more useful to determine the axis of rotation of a general rotation directly. For now, we shall assume that the axis of rotation does actually exist (barring a specific edge case). Any point that lies on this axis should be invariant under the rotation, and so we should have an eigenvalue of 1 for every rotation matrix $\mathbf{R}$.
- In fact, we will formally prove this on the next page.

General 3D Rotations

- Notice that $\mathbf{R}_x(\theta)$, $\mathbf{R}_y(\theta)$ and $\mathbf{R}_z(\theta)$ are orthogonal matrices. Intuitively, their respective inverses are obtained by replacing θ by $-\theta$, and one can verify that these inverses are the same as the transposes of the original matrices. One can also verify that the determinants of these matrices are all 1.
- Since the product of orthogonal matrices of the same size is also orthogonal, and if $\mathbf{AB}$, $\det(\mathbf{A})$ and $\det(\mathbf{B})$ are well-defined, then $\det(\mathbf{AB}) = \det(\mathbf{A})\det(\mathbf{B})$, then $\mathbf{R}$ is also orthogonal with $\det(\mathbf{R}) = 1$.
- We now use some identities about matrix transposes and determinants, and the fact that $\mathbf{R}$ is of order 3.
- Hence $\det(\mathbf{R} - \mathbf{I}) = \det(\mathbf{R}^T)\det(\mathbf{R} - \mathbf{I}) = \det(\mathbf{I} - \mathbf{R}^T) = \det((\mathbf{I} - \mathbf{R})^T) = \det(\mathbf{I} - \mathbf{R}) = (-1)^3\det(\mathbf{R} - \mathbf{I}) = -\det(\mathbf{R} - \mathbf{I})$.
- Hence we conclude that 1 is an eigenvalue of every 3×3 general rotation matrix $\mathbf{R}$.
- As with 2D rotations, the other two eigenvalues are complex conjugates and their product must be 1.
- Question: Based on the above facts, when do we fail to define a specific axis of rotation?

Determining the axis and angle of general 3D rotations

- To determine the axis of rotation for a 3×3 general rotation matrix $\mathbf{R}$, suppose $\mathbf{R}$ is **not** symmetric. Consider also a vector $\mathbf{x}$ parallel to the axis of rotation, so that $\mathbf{R}\mathbf{x} = \mathbf{R}^T\mathbf{x}$, i.e. $(\mathbf{R} - \mathbf{R}^T)\mathbf{x} = \mathbf{0}$. This system can be solved by normal means.
- Note that $\mathbf{R}$ is symmetric if and only if the angle of rotation is either 0° or 180° (modulo 360°), since these are the only angles that give matrices that are their own inverses.
- In the former case, $\mathbf{R} = \mathbf{I}$ and the axis of rotation is undefined since the dimension of the eigenspace is 3 and not 1.
- In the latter case, this is equivalent to reflection about a line! Either use the formula shown earlier, or observe that $\mathbf{R}(\mathbf{I} + \mathbf{R}) = \mathbf{R} + \mathbf{R}^2 = \mathbf{R} + \mathbf{R}\mathbf{R}^T = \mathbf{I} + \mathbf{R}$, and since the eigenvalues must be 1, -1 and -1 , by rank-nullity theorem, $\text{rank}(\mathbf{I} + \mathbf{R}) = 3 - 2 = 1$. That is, any non-zero column of $\mathbf{I} + \mathbf{R}$ is parallel to the axis of rotation.

Determining the axis and angle of general 3D rotations

- The angle of rotation of $\mathbf{R}$, θ , satisfies $\cos \theta = \frac{\text{tr}(\mathbf{R}) - 1}{2}$ and $\sin \theta = -\frac{\text{tr}(\mathbf{NR})}{2}$, where if $\mathbf{n} = \begin{pmatrix} n_1 \\ n_2 \\ n_3 \end{pmatrix}$ is a unit vector parallel to the axis of rotation, then $\mathbf{N} = \begin{pmatrix} 0 & -n_3 & n_2 \\ n_3 & 0 & -n_1 \\ -n_2 & n_1 & 0 \end{pmatrix}$.
- Details are available in this paper*. (Due to lack of time before the milestone, I have yet to abridge the explanation given in the paper; it essentially involves cross products and the orthogonal projection theorem.)

* <https://arxiv.org/pdf/1810.02999>

Challenge! A result on 3D rotations

Suppose the angle of rotation of a 3D rotation matrix $\mathbf{R}$ is not a multiple of 360° (including 0°). Prove that the axis of rotation exists and can be uniquely determined from the entries of $\mathbf{R}$.

Note that we have previously *assumed* only its existence; we now ask for a *formal proof* of existence **and** uniqueness.

Sketch of Proof

- We first prove the case where the rotation axis passes through the origin, and so $\mathbf{R}$ can be written as a 3×3 matrix.
- Firstly, 1 is an eigenvalue of every 3×3 rotation matrix (see earlier proof). Since the angle of rotation is not a multiple of 360° (including 0°), $\mathbf{R} \neq \mathbf{I}$, so the dimension of the column space (or row space) of $\mathbf{R} - \mathbf{I}$ is at least 1. That is, the geometric multiplicity of the eigenvalue 1 is at most 2.
- However, a geometric multiplicity of 2 would imply an algebraic multiplicity of at least 2. This is impossible, as:
 - The product of eigenvalues of $\mathbf{R}$ is 1 since $\mathbf{R}$ is orthogonal. An algebraic multiplicity of 2 leads to the third eigenvalue also being 1, which is a contradiction.
 - An algebraic multiplicity of 3 implies that the characteristic polynomial of $\mathbf{R}$ is $(\lambda - 1)^3$, and so $\text{tr}(\mathbf{R}) = 3$ by Vieta's formula. But since $\mathbf{R}$ is orthogonal, all rows and columns must be unit vectors. The only possibility is that $\mathbf{R} = \mathbf{I}$; contradiction.
- Hence, the geometric multiplicity of 1 must be 1, that is, the dimension of its corresponding eigenspace is 1. Geometrically, it represents a line containing all points invariant under the rotation, i.e. the axis of rotation exists. Uniqueness is trivial.
- In the 4×4 matrix case (for affine rotations), we are also guaranteed one and only one axis so long as $\mathbf{R}$ is not a translation matrix. This time, we solve the system $\mathbf{R}\mathbf{x} + \mathbf{t} = \mathbf{x}$, i.e. $(\mathbf{I} - \mathbf{R})\mathbf{x} = \mathbf{t}$. The fourth row of $\mathbf{R}$ is $(0 \ 0 \ 0 \ 1)$, and the fourth entries of $\mathbf{x}$ and $\mathbf{t}$ are both 1, so a solution for $\mathbf{x}$ exists.
- For uniqueness, suppose that two different axes existed. Then, their direction vectors must both lie in $\ker(\mathbf{I} - \mathbf{R})$. Then, the set of all points invariant under the rotation would be a subspace of dimension 2 (or a plane), which is a contradiction.

So how does Visu(AL)-I actually work?

- Firstly, every object has a set of vertices in 3D, and a ‘matrix stack’ that initially only contains the identity matrix. Its ‘current matrix’ is also the identity matrix.
- The formulas for converting translations, reflections, rotations and the like are converted to matrix form, and the ‘matrix stack’ and ‘current matrix’ are updated accordingly.
- The new transformation is applied to the object vertex-wise. Once the object’s vertex coordinates have been updated, the changes are displayed on the OpenGL panel.
- Another object is also created upon the first transformation of an object – the ‘shadow object’. It represents the last position of the current object, and is drawn as a translucent object. Subsequently, the ‘shadow object’ gets updated to the current object prior to the most recent transformation.
- Corresponding vertices between the current object and the ‘shadow object’ are connected with translucent edges, to show the exact movement of each vertex.

A Caveat in Chaining Transformations

- Consider maintaining a stack of transformation matrices for transforming a space in $\mathbb{R}^n$. In fact, VISU(AL)-1 does this for most matrices in the $\mathbb{R}^3$ case. Consider the following operations:
 - A variable to hold the current matrix $\mathbf{C}$, a result of pre-multiplying the 3×3 transformation matrices with $\mathbf{I}_3$, starting with the matrices inserted earlier (from the bottom of the stack).
 - A push operation which pushes a new transformation matrix $\mathbf{M}$ onto the stack. Then, update $\mathbf{C}_{after} = \mathbf{M}\mathbf{C}_{before}$.
 - A pop operation which works as follows: It peeks the top of the stack to obtain the most recently pushed transformation matrix $\mathbf{M}$, and then pushes $\mathbf{M}^{-1}$ into the stack using the above-described push operation, so that $\mathbf{C}_{after} = \mathbf{M}^{-1}\mathbf{M}\mathbf{C}_{before} = \mathbf{C}_{before}$. Then, the top two matrices in the stack (which would be $\mathbf{M}^{-1}$ and $\mathbf{M}$) are popped.
- The above stack provides convenience in implementing an ‘undo’ feature for a computer program. However, this assumes every transformation matrix pushed into the stack is invertible, which may not be the case!
- Can you give an example of a non-trivial linear transformation which cannot be reversed? That is, its standard matrix is singular, but it is not the zero matrix. Geometrically, why is this transformation irreversible?

One possible answer: Consider the transformation matrix $\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$ which projects any vector in $\mathbb{R}^2$ onto the x-axis. By projecting a vector onto a subspace of a smaller dimension, we lose at least one dimension of information that cannot be recovered, since different vectors in $\mathbb{R}^n$ can be projected to the same vector in a given subspace of $\mathbb{R}^n$ with a smaller dimension. Note that homogenous coordinates cannot save us here either! Try to think of why this is so.

The Resolution

- A second stack is used to hold ‘old current matrices’.
- Transformation matrices which are singular (e.g. projection matrices) are pushed onto the main stack as usual.
- If $\mathbf{M}$ is singular, before updating $\mathbf{C}_{after} = \mathbf{M}\mathbf{C}_{before}$, push $\mathbf{C}_{before}$ onto the second stack.
- When we want to pop a matrix $\mathbf{M}_{top}$ from the main stack, we peek the matrix at the top of the stack and its determinant is calculated.
 - If $\det(\mathbf{M}_{top})$ is not close to 0, then $\mathbf{M}_{top}$ is invertible and the previously described algorithm is applied.
 - If $\det(\mathbf{M}_{top})$ is close to 0, then $\mathbf{M}_{top}$ represents a non-invertible transformation. There should exist a matrix at the top of the second stack, which must be the ‘old current matrix’ right before $\mathbf{M}_{top}$ was pushed onto the main stack. We pop the ‘old current matrix’ from the second stack and restore it as the ‘present current matrix’, then pop $\mathbf{M}_{top}$ from the main stack.
- Although the two-stack solution also works for invertible matrices, assuming that invertible transformations are more common than non-invertible ones, this hybrid solution is more efficient memory-wise.